TARUN SINGH

PhD Scholar, Department of Civil Engineering, Indian Institute of Technology Roorkee

Google Scholar | ORCID | LinkedIn | Academic webpage

RESEARCH PROFILE

PhD researcher working on constitutive modelling of rate-dependent materials, with emphasis on finite-deformation continuum mechanics, rheological connections, evolving natural configurations, and thermodynamically consistent modelling of viscoelastic solids.

EDUCATION

PhD in Civil Engineering 2023–Present
Indian Institute of Technology Roorkee
Research: Constitutive modelling of rate-dependent materials | Supervisor: Dr. Sandipan Paul

M.Tech. in Structural Engineering 2020–2022
Dr. B. R. Ambedkar National Institute of Technology Jalandhar
Thesis: Carbonation resistance of concrete made with iron slag and recycled concrete aggregate

B.Tech. in Civil Engineering 2015–2019
Rewa Engineering College, Rewa, Madhya Pradesh

RESEARCH EXPERIENCE

- PhD research on constitutive modelling of rate-dependent materials and finite-deformation viscoelasticity.
- Development of constitutive frameworks using rheological connections and multiple/evolving natural configurations.
- Formulation and numerical integration of nonlinear constitutive equations, with parameter identification using experimental response data.
- M.Tech. research on durability and performance of concrete incorporating iron slag and recycled concrete aggregates.

JOURNAL PUBLICATIONS

1. **T. Singh** and S. Paul (2026). *"A deformation mode-specific constitutive framework for finite viscoelasticity based on scalar, conjugate stress/strain pairs: Applications to characterization of soft materials."* Journal of the Mechanics and Physics of Solids, Article 106895. DOI: https://doi.org/10.1016/j.jmps.2026.106895
2. **T. Singh** and S. Paul (2026). *"Constitutive modeling of Kelvin–Voigt type viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations."* International Journal of Solids and Structures, 341, Article 114336. DOI: https://doi.org/10.1016/j.ijsolstr.2026.114336
3. **T. Singh** and S. Paul (2025). *"On the extension of the concept of rheological connections to a finite deformation framework using multiple natural configurations."* International Journal of Engineering Science, 212, Article 104267. DOI: https://doi.org/10.1016/j.ijengsci.2025.104267
4. N. Singh, A. Singh, N. Ankur, P. Kumar, M. Kumar, and **T. Singh** (2022). *"Reviewing the properties of recycled concrete aggregates and iron slag in concrete."* Journal of Building Engineering, 60, 105150.
5. N. Singh, N. Ankur, **T. Singh**, M. Kumar, A. Singh, and P. Kumar (2022). *"Non-destructive evaluation of iron slag concrete containing recycled concrete aggregates."* Indian Concrete Journal, 96(11).
6. N. Singh, **T. Singh**, M. Kumar, A. Singh, and P. Kumar (2022). *"Investigating the fresh state performance of concrete containing iron slag and recycled concrete aggregates."* Materials Today: Proceedings, 65, 1467–1477.

CONFERENCE PRESENTATIONS

1. T. Singh and S. Paul (2026). *"Constitutive modeling of viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations."* IUTAM Symposium: Rate Dependent Mechanics and Multiscale Modelling of Materials, IIT Delhi.
2. T. Singh and S. Paul (2026). *"Thermodynamically consistent constitutive model for viscoelastic solids under finite deformation using evolving natural configurations."* 7th Indian Conference on Applied Mechanics, IIT Kanpur.
3. T. Singh and S. Paul (2025). *"Challenges in extending rheological connections to a finite deformation theory."* 14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), IIT Roorkee.

TECHNICAL SKILLS

Scientific computing: MATLAB

Finite-element and constitutive implementation: Abaqus, UMAT, Fortran, FEniCSx

Numerical methods: Numerical integration of constitutive equations, parameter identification, and model calibration